



NATIONAL SCHOOL OF BUSINESS MANAGEMENT

Model Paper

Year -- Semester -- Examination
SE102.3 - Web Application Development

Instructions to Candidates

- 1) **Answer all the questions.**
- 2) **Mark answers for Part (A) in the provided answering grid. [Color the box of the correct answer.]**
- 3) **Write answers for all the questions in Part (B) in the provided booklet.**
- 4) **Part (A) carries 25 marks and Part (B) carries 75 Marks.**
- 5) **Illustrate answers with sketches where required.**
- 6) **Time allocated for the examination is three (03) hours.**
- 7) **If a page or a part of this question paper is not printed, please inform the Supervisor immediately.**
- 8) **Write your index number in all pages of answer script.**
- 9) **Attach all answer sheets at the end of the examination.**

Part A

1. What is the best practice for ensuring a consistent layout across multiple web pages?
 - A) Using inline styles for each element
 - B) Defining styles in an external CSS file
 - C) Hardcoding all styles in each HTML file
 - D) Using randomly selected fonts and colors
2. Which of the following is a valid way of declaring a variable in PHP?
 - A) \$variable123
 - B) 123_variable
 - C) -variable
 - D) !variable
3. Which HTML tag is used to include an audio file?
 - A) <sound src="audio.mp3"></sound>
 - B) <audio controls><source src="audio.mp3" type="audio/mpeg"></audio>
 - C) <media src="audio.mp3"></media>
 - D) <track src="audio.mp3"></track>
4. Which of the following is NOT a valid JavaScript variable name?
 - A) \$myName
 - B) my_name
 - C) 5name
 - D) _name5
5. What is the purpose of the "for" loop in JavaScript?
 - A) Declare a function
 - B) Iterate over a block of code a fixed number of times
 - C) Handle errors in code
 - D) Terminate script execution
6. Which of the following is a good practice for designing user-friendly forms on a website
 - A) Including unnecessary fields
 - B) Grouping related fields together
 - C) Using only placeholder text without labels
 - D) Using very small input boxes
7. Which of the following provides the code changing size of the paragraph text size?
 - A) <p size="14px">
 - B) <p style="font-size: 14px;">
 - C) <p style="text-size: 14px;">
 - D) <p font-size="14px;">
8. Which of the following is the correct way to print a message in the browser's console using JavaScript?
 - A) console.print("Hello, World!");
 - B) print("Hello, World!");
 - C) console.log("Hello, World!");
 - D) log.console("Hello, World!");
9. How do you set the background color of a HTML webpage using inline CSS?
 - A) <body style="background-color: red;">
 - B) <html style="background-color: red">
 - C) <page style="background-color: blue;">
 - D) <bg style="background-color: blue;">

10. What will be the output of the php code snippet ?

```
function sample() {  
    $x = 5;  
    return $x + 10;  
}  
$x = 20;  
echo sample();
```

A) error

B) 20

C) 15

D) testScope()

11. What is the output of the following code?

```
var a = "10";  
var b = 2;  
var c = a / b;
```

A) 10/2

B) 102

C) 5

D) none of the above

12. What is the correct HTML for creating a hyperlink?

A) wmc

C) <a> http://www.yy.com

B) wmc

D) wmc

13. What is the correct syntax to add a new HTML element dynamically inside a div with an ID of "container"? <div id="container"></div>

A) document.createElement("div").innerHTML = "New Element";

B) document.getElementById("container").innerHTML = "<div>New Element</div>";

C) document.getElementsByName("container").innerHTML = "<div>New Element</div>";

D) document.getElement("container").innerHTML = "<div>New Element</div>";

14. What is the correct HTML for inserting an image?

A) <imges src="image.jpg" alt="MyImage">

C)

B)

D) <images alt="image.jpg" src = "MImage">

15. HTML stands for,

A) HighText Machine Language

C) HyperText Machine Language

B) HyperText and links Markup Language

D) HyperText Markup Language

16. What is the correct way to include an external JavaScript file in HTML?

A) <script href="script.js"></script>

C) <scriptlink="script.js"></script>

B) <script src="script.js"></script>

D) <script file="script.js"></script>

17. What is the primary goal of responsive web design?

A) To create websites with flashy animations

computers

B) To ensure websites look good on all devices and screen sizes

D) To use only fixed-width layouts

C) To make websites load faster on desktop

18. Which of the following method can be used to close a MySQL database using PHP?

A) mysql_connect()

B) mysql_query()

C) mysql_close()

D) None of the above

19. What is the output of this JavaScript Snippet?

```
<html><script>
    function showMessage(message) {
        alert(message);
        return message.length;    }
</script><body>
<button onclick="let length = showMessage('Hello, world!');
console.log(length);">Show Message </button>
<body></html>
```

- A) Displays an alert with "Hello, World!" and logs 13 to the console
- B) Logs 12 to the console without displaying an alert
- C) Displays an alert with "Hello, World!" and logs 12 to the console
- D) Logs 13 to the console without displaying an alert.

20. What will be the output of the following PHP code?

```
function multiply($a, $b = 2) {
    return $a * $b;
}
echo multiply(5);
```

- A) 7
- B) 10
- C) 25
- D) Error

21. Which CSS property can be used to control the space between the content and the border of an element?

- A) margin
- B) padding
- C) border-spacing
- D) line-height

22. Which HTML5 element is used to Which of the following is the correct syntax for an if statement in JavaScript?

- A) if x = 5 { console.log("Hello"); }
- B) if (x == 5) { console.log("Hello"); }
- C) if x == 5 then { console.log("Hello"); }
- D) if (x == 5) console.log("Hello"); else;

23. Which PHP superglobal is used to collect form data sent with both GET and POST methods?

- A) \$_POST
- B) \$_GET
- C) \$_REQUEST
- D) \$_SESSION

24. What is the correct syntax to declare a constant in JavaScript?

- A) let PI = 3.14;
- B) const PI = 3.14;
- C) var PI = 3.14;
- D) constant PI = 3.14;

25. What will be the output of the following PHP Code?

```
function operation($x, $y, $op) {
    switch ($op) {
        case 'subtract':
            return $x - $y;
        case 'divide':
            return $x / $y;
        default:
            return 'Invalid operation';
    }
}
echo operation(20, 5, 'subtract');
```

A) 5

B) 15

C) 50

D) Invalid operation

Part A – Answering Grid

[Instruction: Color the box of the correct answer.]

Example	a	b	c	d
1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
5	a	b	c	d
6	a	b	c	d
7	a	b	c	d
8	a	b	c	d
9	a	b	c	d
10	a	b	c	d
11	a	b	c	d
12	a	b	c	d
13	a	b	c	d
14	a	b	c	d
15	a	b	c	d
16	a	b	c	d
17	a	b	c	d
18	a	b	c	d
19	a	b	c	d
20	a	b	c	d
21	a	b	c	d
22	a	b	c	d
23	a	b	c	d
24	a	b	c	d
25	a	b	c	d

(1 * 25 = 25 Marks)

PART A

19.

```
<html><script>
    function showMessage(message) {
        alert(message);
        return message.length;    }
</script><body>
<button onclick="let length = showMessage('Hello, world!');
console.log(length);">Show Message </button>
<body></html>
```

Part B

Question 01

[25 Marks]

- (a) Write HTML code integrated with inline CSS to create the table below. Write the JavaScript function to display the alert message as “Price of this course is \$200” once the button clicked.

Follow the main instructions; other design elements can be implemented by observing the output.

- Create a hyperlink as "LearningHub.com" linking to "https://www.learning.com".
- Insert an image named "education.jpg" with dimensions width 200px by height 170px.

[12 Marks]

AI Learning Hub Courses

Learning Hub	Course Details		
	Course and Link	Details	Price
	Basics of Deep learning - LearningHub.com	Our Core Focus: Machine Learning Algorithms	<button>Show Price</button>

- (b) Write the HTML code for the form in concert.html. Write the PHP code in concert_display.php to retrieve and display the entered details after submission.

[13 Marks]

Concert Registration Form

Username:

Password:

Email:

Ticket Type: ☐ VIP ☐ Regular

Registration Details

Username: Shevin De Silva

Password: 12345

Email: shevin@gmail.com

Ticket Type: Regular

Question 02

[25 Marks]

- a) Write the **JavaScript Function**. This web page should allow the user to select the number of dishes ordered from a dropdown list (e.g., Pizza - \$15, Burger - \$10, Pasta - \$12) and enter the quantity for each dish. Calculate the total bill. Additionally, apply a 10% discount if the total amount exceeds \$50. **(Note: No need for HTML/CSS codes to create this form; focus on functionality with JavaScript.)** Use the formula: Total Bill = (Price per Dish * Quantity) - Discount (if applicable)

Restaurant Bill Calculator

Select a Dish:

Quantity:

Total Bill: \$648.00

[10 marks]

- b) Write the **JavaScript Function** to calculate the total cost of a meal based on the number of items the customer orders.
If they want to add a dessert, they can check a box to include it for an extra cost. If the customer orders 3 or fewer items, the meal costs \$15. If they order more than 3 items, the meal costs \$20. If they check the box for dessert, add \$5 to the total.

You also need to make sure to apply **JavaScript validation** to users enter a valid number of items (greater than 0) and check the dessert box if they want the dessert. (No need of HTML/ CSS Code)

Meal Ordering System
How many items do you want to order?
Want a dessert for \$5? ☐

[10 marks]

- c) Here is the book purchase calculator form that allows users to enter the number of books they want to purchase.

Write a **PHP script** to calculate the total cost based on the fixed book price and the number of books entered by the user. (No need of HTML/CSS Code) [5 Marks]

Book Purchase Calculator
Type of Book Available : Science Friction
Fixed Price of a book : \$20
Number of Books:

Question 03

[25 Marks]

Use the knowledge gained from your group project to answer the following questions. Short answers are preferred. Any assumptions can be made related to this scenario.

Scenario:

PizzaPals, a pizza shop, operates a web application that many users rely on for ordering pizza. Its key features include selecting pizzas, placing orders with online payments, and using a delivery service. However, due to issues with the UI design, customers face challenges navigating the system. Your goal is to redesign the system to make it more user-friendly and improve overall experience.

1. Who are the main users of the PizzaPals website?
2. Name three groups of people you could consult to gather ideas for improving the platform.
3. What are the main pages you would include on the redesigned website?
4. List four possible UI-related issues that might make the current website harder for customers to use.
5. Suggest three features you could add to make the website more customer friendly.

6. Name a UI framework you could use to build the front end of the website.
7. What customization options should be included on the product page for pizzas? (List 3)
8. What type of database would you use to store product details, customer information, and order history?
9. Other than a web application, what would be a suitable platform for PizzaPals' online ordering system?
10. What kind of visual or feedback elements (e.g., progress bars, notifications) could be added to improve the order tracking experience? (List 3)
11. List two security features would you implement to protect customer data and ensure secure online payments?
12. Name two emerging technologies that could improve the functionality of PizzaPals' web application.

END